FREE SAMPLE · CHAPTERS 1-3

THE AQUARIUM RESET

Stop losing fish, and build a tank that stays healthy.



About This Sample

This is a free extract from **The Aquarium Reset**, a step-by-step system for keeping freshwater fish healthy.

The three chapters here explain why most beginner fish die, how the nitrogen cycle works (and how to cycle a tank properly), and how to test your water. Get those three right, and most fish deaths stop.

CHAPTER 1

Why Your Fish Keep Dying

Fish rarely die of old age in a beginner's tank. They die of problems in the water that you can't see or smell. You usually don't notice them until fish start gasping, hiding or disappearing.

The good news: almost all of these problems are preventable, and most come down to a handful of causes.

The real reasons fish die

1. The tank wasn't cycled. A new tank has no bacteria to break down fish waste. The waste builds up as ammonia, which burns fish gills and poisons them. This is so common it has a name: **new tank syndrome**. (Chapter 2)

2. Overfeeding. Uneaten food rots and produces even more ammonia. Most beginners feed far more than their fish need. (Chapter 7)

3. Too many fish, too soon. Every fish adds waste. Adding lots of fish at once overwhelms the bacteria that keep the water safe. (Chapter 6)

4. The wrong fish for the tank. Some fish sold to beginners grow to 30 cm (12 in) or more, need large groups, or attack their tankmates. (Chapters 6 and 14)

5. Skipped maintenance. Without regular water changes, waste slowly builds up until fish become stressed and sick. (Chapter 8)

6. Sudden changes. Big swings in temperature or water chemistry shock fish. A common cause is a huge water change with untreated or badly matched water. (Chapter 8)

7. Disease brought in by new fish. New fish can carry parasites and infections into a healthy tank. (Chapters 6 and 10)

Fish live in their own waste

This is the most useful thing to understand about aquariums.

In a river or lake, fish waste is diluted in huge volumes of moving water. In a tank, fish eat, breathe and produce waste in one small, closed box of water. **Everything that goes into the tank stays in the tank** until bacteria break it down or you remove it with a water change.

That's why the most important piece of equipment isn't the filter or the heater. It's the test kit.

Clear water isn't clean water

Ammonia and nitrite, the two substances that kill most beginner fish, are completely invisible. A tank can look crystal clear and still be toxic. The only way to know is to test.

> **The rule for this whole guide:** when something looks wrong, test the water before you do anything else.

Stress is the hidden link

Poor water quality rarely kills fish instantly. More often it stresses them, and stressed fish have weaker immune systems. That's why a tank with an ammonia problem so often ends up with a disease outbreak too.

Fix the water, and many "disease" problems never start, or clear up much more easily.

CHAPTER 2

The Nitrogen Cycle

If you only read one chapter of this guide, make it this one.

What the cycle is

Fish give off **ammonia** through their gills and in their waste. Rotting food and decaying plants produce it too. Ammonia is highly toxic to fish.

In an established tank, two groups of beneficial bacteria deal with it, and you do the last step:

1. The first group turns **ammonia** into **nitrite**. Nitrite is also toxic.
2. The second group turns **nitrite** into **nitrate**. Nitrate is far less harmful, but it builds up over time.
3. **You** remove nitrate with regular water changes. Live plants also use some of it.

Stage	Toxic to fish?	What removes it
Ammonia	Yes, highly	Bacteria turn it into nitrite
Nitrite	Yes, highly	Bacteria turn it into nitrate
Nitrate	Only at high levels	Water changes and live plants

A **cycled tank** is simply a tank with enough of these bacteria to process all the ammonia and nitrite as fast as they're produced.

Where the bacteria live

Beneficial bacteria don't float in the water. They live on surfaces, and above all **in your filter media**: the sponges, ceramic rings or other material inside the filter. They also grow on gravel, decorations and glass.

That means three common habits can wipe them out:

- **Rinsing filter media under the tap.** Chlorine and chloramine in tap water are there to kill bacteria, and they kill these ones too. (Chapter 8)
- **Replacing all the filter media at once.** This throws most of the bacteria away.
- **Switching the filter off** for more than a few hours. Without flowing water, the bacteria run out of oxygen.

Do any of these, and a cycled tank can "crash" back into an ammonia problem.

How long it takes

Cycling a new tank from scratch usually takes **4 to 8 weeks**, sometimes longer. You can't really rush it. Bottled bacteria and filter media from an established tank can help it along.

That wait is the hardest part of starting an aquarium, and it's why so many beginners skip it. **Don't skip it.**

Option 1: Fishless cycling (recommended)

You cycle the tank before any fish go in, so no fish are harmed while you wait.

What you need:

- a fully set-up tank with the filter and heater running
- a water conditioner
- a liquid test kit (Chapter 3)
- a source of ammonia: pure ammonia sold for fishless cycling is easiest. Fish food also works, but it's slower and messier.

1. **Set up the tank fully** (Chapter 5). Treat the water with conditioner, and run the filter and heater. Warmer water, around 26–28°C (79–82°F), helps the bacteria grow faster.
2. **Add ammonia** until your test reads about **2 ppm**. Note how much you added.
3. **Add bottled bacteria** if you're using it, following the label.
4. **Test every day or two** for ammonia, nitrite and nitrate. Record the results in the **Cycle Tracker (Appendix A)**.
5. **Watch for ammonia falling and nitrite appearing.** That means the first bacteria have arrived. Each time ammonia drops close to zero, top it back up to about 2 ppm.
6. **Watch for nitrite falling and nitrate appearing.** That means the second bacteria have arrived. Keep adding ammonia to feed both groups.
7. **Check whether the tank is cycled.** Add ammonia to 2 ppm. If **both ammonia and nitrite read 0 within 24 hours**, with nitrate present, it's done. Confirm this two or three days in a row.
8. **Do a large water change** of 50–75% or more. This brings nitrate down and lets you return the temperature to your fish's range. Add your first fish within a day or two, so the bacteria don't starve.

> **If nitrite stays very high for weeks and nothing changes,** do a partial water change to bring it down. Very high nitrite can slow the bacteria you're trying to grow.

Option 2: Fish-in cycling (if fish are already in the tank)

If you already have fish in an uncycled tank, don't panic, and don't take them back. You can cycle with fish in the tank, but you need to protect them from ammonia and nitrite while the bacteria grow.

1. **Test every day** for ammonia and nitrite.
2. **If either reads 0.5 ppm or higher, change 30–50% of the water.** Use conditioned water at the same temperature as the tank. Keep both readings as close to zero as you can.
3. **Use a water conditioner that says it detoxifies ammonia and nitrite.** It gives fish some temporary protection between water changes.
4. **Feed very lightly:** once a day, or every other day. Less food means less ammonia.
5. **Don't add any more fish** until the tank is fully cycled.
6. **Don't clean the filter** during this time.
7. **Keep going** until ammonia and nitrite stay at 0 for at least a week without water changes, and nitrate is rising.

Warning signs of a cycle problem

- Fish gasping at the surface
- Red or inflamed gills
- Fish lying on the bottom, or holding their fins clamped to their body
- Sudden deaths in a new tank, or after the filter was cleaned or replaced

If you see any of these, test immediately and follow the rescue steps in **Chapter 11**.

CHAPTER 3

Testing Your Water

Testing turns guessing into knowing. It takes about ten minutes. More than any other habit, it separates people who keep losing fish from people who don't.

Liquid kits vs test strips

Liquid test kits use drops of reagent in small test tubes. They're more accurate, and cheaper per test in the long run. **This is what you want**, especially while cycling.

Test strips are quick and convenient, but generally less accurate, especially for ammonia. They're fine for a quick check on an established tank. Don't rely on them to tell you whether a tank is cycled.

For accurate results:

- Follow the instructions exactly, including wait times and shaking the bottles.
- Read the colours in good natural light, against the white part of the colour card.
- Check expiry dates. Old reagents give false readings.
- Rinse the test tubes in tank water. Never use soap.

What to test

- **The core three:** ammonia, nitrite and nitrate.
- **The next most useful:** pH and temperature.
- **Once at set-up, and occasionally after:** GH and KH (explained below).

Target numbers

Test	Target for most tropical community fish
Ammonia	0 ppm
Nitrite	0 ppm
Nitrate	Ideally below 20 ppm, and no higher than 40 ppm
pH	Stable, usually between 6.5 and 7.8
Temperature	24–26°C (75–79°F) for most species

In an established tank, any ammonia or nitrite reading above 0 is a warning sign. Find out why. (Chapter 9)

Why ammonia is worse at higher pH

Test kits measure total ammonia, but not all of it is equally dangerous. In warmer water with a higher pH, more of it takes its most toxic form.

So 0.5 ppm is more dangerous in a warm tank at pH 8 than in a cooler tank at pH 6.5. Either way, the target is zero.

pH: stability matters more than the number

Most common aquarium fish adapt well to a wide range of pH, especially captive-bred ones. What harms them is **sudden change**.

This leads to a very common beginner mistake: **chasing the "perfect" pH** with pH-up and pH-down chemicals. The pH drifts back, more chemicals go in, and the fish are stressed by the constant swings.

Unless you keep a species with very specific needs, **choose fish that suit your tap water** instead of fighting it.

GH and KH, simply

GH (general hardness) is roughly how much calcium and magnesium is in the water. "Hard water" has a high GH, "soft water" a low one. Some fish prefer soft water, others hard.

KH (carbonate hardness) is how well the water resists changes in pH. Water with very low KH can suffer sudden pH crashes, especially in older tanks that go too long without water changes.

You don't need to master these. Know your numbers, and use them to choose suitable fish.

Test your tap water too

Test your tap water once for ammonia, nitrite, nitrate, pH, GH and KH. Some tap water contains measurable ammonia or nitrate, which changes how you read your tank results.

Find out whether your water supplier uses **chlorine or chloramine**. Most suppliers publish this online.

- Letting water stand overnight can remove some chlorine.
- It does **not** remove chloramine.
- The simple, reliable answer is a water conditioner that treats both. Use it every time you add tap water.

How often to test

Situation	What to test	How often
Cycling a new tank	Ammonia, nitrite, nitrate	Every 1–2 days

Situation	What to test	How often
First 3 months with fish	Ammonia, nitrite, nitrate, pH	Weekly, before your water change
Established tank	Ammonia, nitrite, nitrate, pH	Every 1–2 weeks
After adding fish, or when something looks wrong	Everything	Straight away

Record your results in the **Water Test Log (Appendix B)**. Patterns over several weeks tell you far more than any single reading.

What's In The Full Guide

This sample covered the first three chapters. The complete guide continues with:

- **Choosing your tank and equipment**, and why bigger is easier
- **Setting up, step by step**
- **Stocking without crashing your tank**: compatibility, fish to avoid, buying healthy fish and acclimation
- **Feeding and the maintenance routine**
- **The diagnostic**: what gasping, clamped fins, flashing and cloudy water actually mean
- **Common diseases**: ich, fin rot, fungus, columnaris, velvet, swim bladder, dropsy and popeye
- **Emergency: ammonia spike rescue**, a step-by-step protocol
- **Algae without panic** and **live plants for beginners**
- **Twelve species profiles**, from bettas and neon tetras to corydoras and cherry shrimp
- **Holidays, moving and power cuts**
- **The ten most common mistakes**
- **A 30-day aquarium reset plan**

Printable extras: cycle tracker, water test and maintenance log, diagnostic cheat sheet, and stocking planner.